

Answer the following shortly:

- 1) Why are AJAX rich applications more vulnerable to XSS attacks?
- 2) Why quick and on demand SQL writing are prone to SQL injection?
- 3) Internet is predominated by world wide web (WWW). Write two applications or protocols that are not www.
- 4) How is a URL resolved by the browser?
- 5) Can we write a code and generate executable without an IDE? Explain

Answers

- **AJAX and XSS Vulnerability:**
 - AJAX allows asynchronous communication with the server, making it possible to fetch and display data dynamically. However, improper handling of user input in AJAX requests can lead to Cross-Site Scripting (XSS) vulnerabilities, where malicious scripts are injected into web pages.
- **Quick and On-Demand SQL Writing and SQL Injection:**
 - Writing SQL queries directly in an application without proper input validation opens the door to SQL injection attacks. Attackers can manipulate input fields to inject malicious SQL code, potentially leading to unauthorized access or data manipulation.
- **Non-WWW Applications/Protocols:**
 - a. **FTP (File Transfer Protocol):** Used for transferring files between a client and a server.
 - b. **SMTP (Simple Mail Transfer Protocol):** Manages the sending of emails over the internet.
- **URL Resolution by the Browser:**
 - When a user enters a URL, the browser resolves it through the Domain Name System (DNS), obtaining the IP address of the server. It then establishes a TCP connection to the server and sends an HTTP request, after which the server responds with the requested web page.
- **Code Generation Without an IDE:**
 - Yes, code can be written and compiled without an Integrated Development Environment (IDE). A simple text editor, command-line tools, and a compiler can be used. Developers write code in a text editor, save the file with the appropriate extension, and use a compiler or interpreter to generate an executable or run the code. IDEs provide a convenient environment with features like code completion and debugging, but they are not strictly necessary for code execution.